# Stochastic Financial Models

Part II

Recorded from the lecturer in 2026-2027.

## 1 Martingales

We work with a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

### 1.1 Definitions

A *filtration* is a sequence of $\sigma$-algebras $(\mathcal{F}_n)_{n\geq 0}$ such that

$$\mathcal{F}_n \subseteq \mathcal{F}_{n+1} \subseteq \mathcal{F}.$$

A *random process* is a sequence $X = (X_n)_{n\geq 0}$ of random variables.
Given a filtration $(\mathcal{F}_n)_{n\geq 0}$ on $(\Omega, \mathcal{F}, \mathbb{P})$, we say that $X$ is a *martingale* if:

(a) $X$ is adapted, i.e. $X_n$ is $\mathcal{F}_n$-measurable for all $n$.

(b) $X$ is integrable, i.e. $X_n$ is integrable for all $n$:

$$\mathbb{E}[|X_n|] < \infty.$$

(c) $X$ satisfies the martingale property:

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n \qquad \text{a.s., for all } n.$$

Given (a) and (b), condition (c) is equivalent to

$$\mathbb{E}[X_{n+1}\mathbf{1}_A] = \mathbb{E}[X_n\mathbf{1}_A] \qquad \text{for all } n \text{ and all } A \in \mathcal{F}_n.$$

If (c) is replaced by

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \geq X_n \qquad \text{a.s., for all } n,$$

or, equivalently,

$$\mathbb{E}[X_{n+1}\mathbf{1}_A] \geq \mathbb{E}[X_n\mathbf{1}_A] \qquad \text{for all } n \text{ and all } A \in \mathcal{F}_n,$$

then $(X_n)_{n\geq 0}$ is a *submartingale*.

If (c) is replaced by

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \leq X_n \qquad \text{a.s., for all } n,$$

or, equivalently,

$$\mathbb{E}[X_{n+1}\mathbf{1}_A] \leq \mathbb{E}[X_n\mathbf{1}_A] \qquad \text{for all } n \text{ and all } A \in \mathcal{F}_n,$$

then $(X_n)_{n\geq 0}$ is a *supermartingale*.

### 1.2 Natural filtration

Given a random process $X = (X_n)_{n\geq 0}$, its *natural filtration* $(\mathcal{F}_n^X)_{n\geq 0}$ is defined by

$$\begin{aligned}\mathcal{F}_n^X &= \sigma(X_0, X_1, \ldots, X_n)\\ &= \left\{\{(X_0, X_1, \ldots, X_n) \in B\} : B \in \mathcal{B}(\mathbb{R}^{n+1})\right\}.\end{aligned}$$

This is the smallest filtration making $X$ adapted.

**Exercise.** Show that a martingale $X$ with respect to $(\mathcal{F}_n)_{n\geq 0}$ is also a martingale with respect to $(\mathcal{F}_n^X)_{n\geq 0}$.

### 1.3 Examples

Let $X_1, X_2, \ldots$ be a sequence of independent integrable random variables. Set

$$\mathcal{F}_n = \mathcal{F}_n^X.$$

Define

$$S_0 = 0, \qquad Z_0 = 1,$$

and

$$S_n = X_1 + \cdots + X_n, \qquad Z_n = \prod_{i=1}^{n} X_i.$$

- If $\mathbb{E}[X_n] = 0$ for all $n$, then $(S_n)_{n\geq 0}$ is a martingale.
- If $X_n \geq 0$ and $\mathbb{E}[X_n] = 1$ for all $n$, then $(Z_n)_{n\geq 0}$ is a martingale.

  To see this, take $A \in \mathcal{F}_n$. Then

$$\begin{aligned}\mathbb{E}[S_{n+1}\mathbf{1}_A] &= \mathbb{E}[X_{n+1}\mathbf{1}_A] + \mathbb{E}[S_n\mathbf{1}_A]\\ &= \underbrace{\mathbb{E}[X_{n+1}]\mathbb{P}(A)}_{=0} + \mathbb{E}[S_n\mathbf{1}_A]\\ &= \mathbb{E}[S_n\mathbf{1}_A].\end{aligned}$$

  Also,

$$\begin{aligned}\mathbb{E}[Z_{n+1}\mathbf{1}_A] &= \mathbb{E}[X_{n+1}Z_n\mathbf{1}_A]\\ &= \underbrace{\mathbb{E}[X_{n+1}]}_{=1}\mathbb{E}[Z_n\mathbf{1}_A]\\ &= \mathbb{E}[Z_n\mathbf{1}_A].\end{aligned}$$

This also shows integrability.

## 2 Conditional Probabilities and Expectation

For $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)},$$

and

$$\mathbb{E}[X \mid B] = \frac{\mathbb{E}[X\mathbf{1}_B]}{\mathbb{P}(B)}.$$

## 2.1 Existence and uniqueness

**Theorem.** Let $\mathcal{G} \subseteq \mathcal{F}$ be a $\sigma$-algebra. Let $X$ be an integrable random variable. Then there exists a random variable $Y$ such that:

(a) $Y$ is $\mathcal{G}$-measurable;

(b)
$$\mathbb{E}[X\mathbf{1}_A] = \mathbb{E}[Y\mathbf{1}_A] \qquad \text{for all } A \in \mathcal{G}.$$

Moreover, if $Y'$ is another random variable satisfying (a) and (b), then

$$Y = Y' \qquad \text{a.s.}$$

**Proof of uniqueness.** Let $X'$ be another integrable random variable with $X \leq X'$. Suppose $Y'$ satisfies (a) and (b) with respect to $X'$.

Set
$$A = \{Y > Y'\}, \qquad Z = (Y - Y')\mathbf{1}_A.$$

Then $A \in \mathcal{G}$ and $Z \geq 0$. Hence

$$\begin{aligned}
\mathbb{E}[Z] &= \mathbb{E}[(Y - Y')\mathbf{1}_A] \\
&= \mathbb{E}[Y\mathbf{1}_A] - \mathbb{E}[Y'\mathbf{1}_A] \\
&= \mathbb{E}[X\mathbf{1}_A] - \mathbb{E}[X'\mathbf{1}_A] \\
&= \mathbb{E}[(X - X')\mathbf{1}_A] \leq 0.
\end{aligned}$$

Therefore $Z = 0$ a.s., so
$$Y \leq Y' \qquad \text{a.s.}$$

Taking $X' = X$, we also obtain $Y \geq Y'$ a.s. Thus

$$Y = Y' \qquad \text{a.s.}$$

## 2.2 Notation

We write
$$Y = \mathbb{E}[X \mid \mathcal{G}] \qquad \text{a.s.}$$

If $\mathcal{G} = \sigma(W)$, where $W$ is a random variable, we write

$$\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X \mid W].$$

If $X = \mathbf{1}_A$, we write
$$\mathbb{E}[X \mid \mathcal{G}] = \mathbb{P}(A \mid \mathcal{G}).$$

**End of Lecture 1**